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Minimax Goodness-of-fit testing in High-Dimensional models

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One can note that since the set of admissible test is too wide, there are no general recommendations as to how to select optimal tests when the alternative is composite and of dimension more than one. A way out of this situation is to characterize the test, not in the usual way by using error functions $\alpha(\psi, P), \beta(\psi, Q)$ for some $P \in \mathcal{P}_0, Q \in \mathcal{P}_1$, but by some monotone functional $F_1(\psi), F_2(\psi)$

2.2.1 General setting

Fix a Borel σ-algebra on the space $\mathcal{P}$ of probability measures (with respect to the total variation distance) such that $\mathcal{P}_0, \mathcal{P}_1$ are Borel subsets. Moreover, fix two priors π_0, π_1 with supports in $\mathcal{P}_0, \mathcal{P}_1$, respectively. Now, define two functionals F_1, F_2 as the average errors with respect to each prior :

$$\begin{aligned} F_1(\psi) &\coloneqq \alpha(\psi, \pi_0) = \mathbb{E}_{\pi_0}\alpha(\psi, P) = \int \alpha(\psi, P) d\pi_0(P) \\ F_2(\psi) &\coloneqq \beta(\psi, \pi_1) = \mathbb{E}_{\pi_1}\beta(\psi, P) = \int \beta(\psi, P) d\pi_1(P) \end{aligned} \tag{2.1}$$

Analogously to what has been done with the standard approach, given $\alpha \in (0, 1)$ we define the set :

$$\Psi_{\alpha,\pi_0} \coloneqq \left\{\psi \in \Psi : \alpha(\psi, \pi_0) \leq \alpha\right\} \tag{2.2}$$

and the following functions :

$$\gamma_t(\psi; \pi_0, \pi_1) \coloneqq t\alpha(\psi, \pi_0) + \beta(\psi, \pi_1)$$

$$\gamma(t; \pi_0, \pi_1) \coloneqq \inf_{\psi \in \Psi} \gamma_t(\psi; \pi_0, \pi_1) \tag{2.3}$$

$$\beta(\alpha; \pi_0, \pi_1) \coloneqq \inf_{\psi \in \Psi_{\alpha,\pi_0}} \beta(\psi, \pi_1) \tag{2.4}$$

The tests $\psi^{(t)}$ or ψ_α are called Bayesian if they provide an infimum eq. (2.3) or in eq. (2.4).

Definition 2.1 (Mixture of a prior). *Given a prior π, one can define its mixture as the measure P_π on $(\mathcal{X}, \mathcal{A})$ given by :*

$$\forall A \in \mathcal{A} : \mathbb{P}_\pi(A) = \mathbb{E}_\pi P(A) = \int_{\mathcal{P}} P(A) d\pi(P)$$

Given all these definitions, the Bayesian hypothesis testing is reduced to testing of the simple hypotheses with null hypothesis $P = P_{\pi_0}$ and alternative $P = P_{\pi_1}$

Remark. *Note that when we have a parametrization which corresponds to an injective map $\Theta \to \mathcal{P}$, we set a Borel σ-algebra $\mathcal{T}$ on Θ (with respect to some topology) that makes the map measurable, one can assume that the priors π_0, π_1 are defined on $(\Theta, \mathcal{T})$. Hence, in our specific context, one can assume that π_0, π_1 are measures on $(\mathbb{R}^n, \mathcal{B}^n)$ in the finite-dimensional case, or on $(\ell^2, \mathcal{B})$ in the infinite-dimensional case (Here $\mathcal{B}$ denotes a Borel σ-algebra on ℓ^2).*

Add details + define the following measure

$$r(dv) = C^{-1}\pi(dv)e^{-\frac{|v|^2}{2}}, C = \int e^{-\frac{|v|^2}{2}}\pi(dv) = L_\pi(0) \tag{2.5}$$

Proposition 2.1. *Let π_0, π_1 be two priors. Then the following equivalence holds :*

$$P_{\pi_0} = P_{\pi_1} \Leftrightarrow \pi_0 = \pi_1$$

Proof. Clearly, if $\pi_0 = \pi_1$, then $P_{\pi_0} = P_{\pi_1}$. Now, assume that $P_{\pi_0} = P_{\pi_1}$ then $L_{\pi_0} = L_{\pi_1}$ almost surely. First, consider the finite-dimensional case $x \in \mathbb{R}^n$. Let μ_1, μ_2 be the probability measures defined in eq. (2.5) for the priors π_0, π_1. By the **previous** proposition, the functions L_{π_0}, L_{π_1} are analytical, this implies that the equality $L_{\pi_0} = L_{\pi_1}$ holds true in the whole space $\mathbb{C}^n$. In particular, we get the equality for $z = ix, x \in \mathbb{R}^n$ which corresponds to the equality of characteristic functions of the probability measures μ_0, μ_1. Therefore $\mu_0 = \mu_1$ which yields $\pi_0 = \pi_1$. In the infinite-dimensional case, the proof above yields equality for all finite dimensional marginals, which implies $\pi_0 = \pi_1$. □

2.3 Minimax approach in Hypothesis testing

2.3.1 General setting

Instead of considering the average error, we consider maximum error, in this section, define :

$$F_1(\psi) = \alpha(\psi, \mathcal{P}_0) = \sup_{P \in \mathcal{P}_0} \alpha(\psi, P) \tag{2.6}$$

$$F_2(\psi) = \beta(\psi, \mathcal{P}_1) = \sup_{P \in \mathcal{P}_1} \beta(\psi, P) \tag{2.7}$$

Moreover, similarly to eq. (2.2), given $\alpha \in (0, 1)$ we define :

$$\Psi_\alpha = \Psi_{\alpha, \mathcal{P}_0} = \left\{\psi \in \Psi : \alpha(\psi, \mathcal{P}_0) \leq \alpha\right\} \tag{2.8}$$

One can also define the following quantities :

$$\gamma_t(\psi, \mathcal{P}_0, \mathcal{P}_1) = t\alpha(\psi, \mathcal{P}_0) + \beta(\psi, \mathcal{P}_1)$$

$$\gamma(t, \mathcal{P}_0, \mathcal{P}_1) = \inf_{\psi \in \Psi} \gamma_t(\psi) \tag{2.9}$$

$$\beta(\alpha, \mathcal{P}_0, \mathcal{P}_1) = \inf_{\psi \in \Psi_{\alpha, \mathcal{P}_0}} \beta(\psi, \mathcal{P}_0, \mathcal{P}_1) \tag{2.10}$$

The tests $\psi^{(t)}$ or ψ_α are called *minimax* if they provide infimums for eq. (2.9) or eq. (2.10).

One can observe the following convexity lemma :

Lemma 2.1. *The following properties hold true :*

1. *The functions* $\alpha(\psi, \mathcal{P}_0), \beta(\psi, \mathcal{P}_1), \gamma_t(\psi, \mathcal{P}_0, \mathcal{P}_1)$ *are convex in* $\psi \in \Psi$
2. *The function* $\alpha \mapsto \beta(\alpha)$ *is convex nonincreasing continuous for* $\alpha \in [0,1]$*, and the function* $t \mapsto \gamma(t)$ *is concave nondecreasing continuous in* $t \in [0,\infty)$*. These two functions are connected by the relations :*

$$\gamma(t) = \inf_{\alpha\in[0,1]} \big(t\alpha + \beta(\alpha)\big) \text{ and } \beta(\alpha) = \sup_{t\in[0,\infty)} \big(\gamma(t) - t\alpha\big) \tag{2.11}$$

Maybe add a proof

2.3.2 Connection with Bayesian approach

Now that the settings of both Minimax and Bayesian approaches were explored, we will see that there exists a close relation between these two. All the following statements go back to Wald [1950]

Definition 2.2 (L_1 distance between two families of measures)**.** *Let* A, B *be two families of measures, we define the* L_1 *distance between* A *and* B *as follows :*

$$d(A, B) := \inf_{P\in A, Q\in B} TV(P, Q)$$

Moreover, we define $[A]$ *to be the convex hull of* A*.*

Theorem 2.1. *Add some text*

1. *The following inequalities hold true :*

$$\gamma(t) \geq \sup\big\{\gamma(t, P_0, P_1) : P_0 \in [\mathcal{P}_0], P_1 \in [\mathcal{P}_1]\big\} \tag{2.12}$$

$$\beta(\alpha) \geq \sup\big\{\beta(\alpha, P_0, P_1) : P_0 \in [\mathcal{P}_0], P_1 \in [\mathcal{P}_1]\big\} \tag{2.13}$$

   In particular, $\gamma = \gamma(1) \geq 1 - \frac{1}{2} d([\mathcal{P}_0], [\mathcal{P}_1])$.

2. *If* $\mathcal{P}_0, \mathcal{P}_1$ *are dominated families of measures on* $(\mathcal{X}, \mathcal{A})$*, and the* $\sigma-$*algebra* $\mathcal{A}$ *is countably generated, then for any* $t > 0, \alpha \in (0,1)$ *the infimas in 2.12 and 2.13 are attained, so the inequalities become equalities.*
3. *If* $\mathcal{P}_0, \mathcal{P}_1$ *are compact with respect to the total varation distance. Then, there exists pairs of priors* $\pi_{0,t}, \pi_{1,t}$ *and* $\pi_{0,\alpha}, \pi_{1,\alpha}$ *such that the minimax tests* $\psi^{(t)}, \psi_\alpha$ *are optimal tests for testing the simple hypothesis* $P_{\pi_{0,t}}$ *or* $P_{\pi_{0,\alpha}}$ *against the simple alternative* $P_{\pi_{1,t}}$*, or* $P_{\pi_{1,\alpha}}$*. The priors* $\pi_{0,t}, \pi_{1,t}$ *and* $\pi_{0,\alpha}, \pi_{1,\alpha}$ *are called least favourable.*

Proof. ADD PROOF □

We will use the previous inequalities 2.12,2.13 in order to construct some lower bounds for the quantities $\gamma(t), \beta(\alpha), \gamma$. For example, let $\pi_{0,n}, \pi_{1,n}$ be sequences of priors such that :

$$\pi_{0,n}(\mathcal{P}_0) = \pi_{1,n}(\mathcal{P}_1) = 1 \tag{2.14}$$

Then, one can rewrite the inequalities as :

$$\gamma(t) \geq \limsup_{n\to\infty} \gamma(t; P_{\pi_{0,n}}, P_{\pi_{0,n}}) \tag{2.15}$$

$$\beta(\alpha) \geq \limsup_{n\to\infty} \beta(\alpha; P_{\pi_{0,n}}, P_{\pi_{1,n}}) \tag{2.16}$$

$$\gamma \geq 1 - \frac{1}{2} \liminf_{n\to\infty} d(P_{\pi_{0,n}}, P_{\pi_{1,n}}) \tag{2.17}$$

One can see that this can be also true if $\pi_{0,n}(\mathcal{P}_0) \to 1$ and $\pi_{1,n}(\mathcal{P}_1) \to 1$

Proposition 2.2. *Let* $\pi_{0,n}, \pi_{1,n}$ *be sequences of priors such that :*

$$\lim_{n\to\infty} \pi_{0,n}(\mathcal{P}_0) = 1, \lim_{n\to\infty} \pi_{1,n}(\mathcal{P}_1) = 1$$

Then, the inequalities 2.15, 2.16 and 2.17 hold true.

Proof. Add proof prop 2.9 IngSus □

Theorem 2.2. *Ingster and Suslina [2012, Theorem 2.3]*
Let π_0, π_1 be priors, and let $S_0 \subseteq \mathcal{P}_0, S_1 \subseteq \mathcal{P}_1$ be measurable sets such that $\pi_0(S_0) = \pi_1(S_1) = 1$. Moreover, assume that the Bayesian tests ψ^t or ψ_α satisfy for $P_0 \in \mathcal{P}_0, P_1 \in \mathcal{P}_1$:

$$\alpha(\psi_\alpha, \mathcal{P}_0) = \alpha(\psi_\alpha, P_0)\ ,\ \beta(\psi_\alpha, \mathcal{P}_1) = \beta(\psi_\alpha, \mathcal{P}_1) \text{ or}$$
$$\gamma_t(\psi^{(t)}; \mathcal{P}_0, \mathcal{P}_1) = \psi(\psi^{(t)}; P_0, P_1)$$

Then, (π_0, π_1) is a pair of least favourable priors under the criteria $\gamma(t)$ or $\beta(\alpha)$, and the tests $\psi^{(t)}$ or ψ_α are minimax.

Proof. Add proof . □

Chapter 3

Minimax distinguishability in the Gaussian sequence model

In what follows, let $(\mathcal{X}, \mathcal{A})$ be a measurable space a consider a family $\left(P_\theta\right)_{\theta\in\Theta}$ of probability measures on the latter measurable space. Let us have an observation $X \in \mathcal{X}$ generated by P_θ for some unknown parameter $\theta \in \Theta$.

3.1 Gaussian sequence model

Definition 3.1. *We observe a sequence $X = (X_i)_{i\in I} \in \ell^2$ which is of the following form :*

$$X = v + \eta \tag{3.1}$$

Here, $v \in \ell^2$ is the (unknown) mean vector, $\eta_i \overset{i.i.d}{\sim} \mathcal{N}(0, 1)$

We denote by P_v the distribution of X when the signal is v, and by P_0 the distribution of X when $v = 0$. Naturally, one can define the following sets :

$$\mathcal{P}_n^G := \left\{P_v, v \in \mathbb{R}^n\right\} \text{ and } \mathcal{P}_\infty^G := \left\{P_v, v \in \ell^2\right\}$$

Remark. *Note that in the case where I is countably infinite, we often assume that $I = \mathbb{N}$, and ℓ^2 is the Hilbert space of square integrable sequences i.e. :*

$$\ell^2 := \left\{(u_i)_{i\in I} : \sum_{i\in I} u_i^2 < \infty\right\}$$

Add more intuition about the choice of the model

3.2 Classical theory of minimax distinguishability

3.2.1 Asymptotic Framework

In this chapter, instead of considering only one hypothesis testing problem, we will follow an asymptotic approach in which we will be considering a family of Hypothesis testing problems indexed by a parameter $\varepsilon \to \varepsilon_0$. Depending on the setting, this parameter may be the dimensions $n \to \infty$, the sample size $N \to \infty$ or the white noise level $\varepsilon \to 0$. More generally, this corresponds to increasing the amount of available information.
Consider an observation $X_\varepsilon \in \mathcal{X}_\varepsilon$ generated by a probability measure $\mathbb{P}_{\varepsilon,\theta}$ on the measurable space $(\mathcal{X}_\varepsilon, \mathcal{A}_\varepsilon)$ where the parameter θ is unknown. Given two families of measurable subsets $\Theta_{\varepsilon,0}, \Theta_{\varepsilon,1}$, we will be testing the following hypotheses :

$$\mathcal{H}_0 : \theta \in \Theta_{\varepsilon,0} \text{ vs } \mathcal{H}_1 : \theta \in \Theta_{\varepsilon,1} \tag{3.2}$$

and we will be studying the performance of tests as $\varepsilon \to \varepsilon_0$. The statistical performance of a family of tests $\{\psi_\varepsilon\}$ can be studied using type I and II errors :

$$\alpha_\varepsilon(\psi_\varepsilon) = \sup_{\theta\in\Theta_{\varepsilon,0}} \mathbb{E}_{\varepsilon,\theta}\psi_\varepsilon \text{ and } \beta_\varepsilon(\psi_\varepsilon) = \sup_{\theta\in\Theta_{\varepsilon,1}} \mathbb{E}_{\varepsilon,\theta}[1-\psi_\varepsilon]$$

Or :

$$\gamma_{\varepsilon,t}(\psi_\varepsilon) = t\alpha_\varepsilon(\psi_\varepsilon) + \beta_\varepsilon(\psi_\varepsilon) \text{ and } \gamma(t) = \inf_{\psi_\varepsilon\in\Psi}\Big(t\alpha_\varepsilon(\psi_\varepsilon) + \beta_\varepsilon(\psi_\varepsilon)\Big)$$

In the minimax setting, we say that a family of tests $\psi_\varepsilon^{(t)}$ is asymptotically minimax is :

$$\gamma_{\varepsilon,t}(\psi_\varepsilon^{(t)}) = \gamma_\varepsilon(t) + o(1) \text{ as } \varepsilon \to \varepsilon_0$$

Under the Neyman-Pearson setting, we consider families of tests ψ_ε satisfying $\alpha_\varepsilon(\psi_\varepsilon) \le \alpha + o(1)$, and we say that a family $\psi_{\varepsilon,\alpha}$ is asymptotically minimax if :

$$\alpha_\varepsilon(\psi_{\varepsilon,\alpha}) \le \alpha + o(1) \text{ and } \beta_\varepsilon(\psi_{\varepsilon,\alpha}) = \beta_\varepsilon(\alpha) + o(1) \text{ as } \varepsilon \to \varepsilon_0$$

We are interested in the study of the asymptotic properties of $\gamma_\varepsilon(t), \beta_\varepsilon(\alpha)$, thanks to Lemma 2.1, this can be done only by studying the asymptotic properties of $\gamma_\varepsilon := \gamma_\varepsilon(1)$. But this kind of problems is really difficult especially in the non parametric case, so we are interested in a simpler kind of problems called *distinguishability problem* (*rate asymptotic problem*) which can be summarized as follows :

- We want to obtain conditions for minimax distinguishability i.e. conditions for $\gamma_\varepsilon \to 0$, and construct *minimax consistent* families of tests ψ_ε such that $\gamma_\varepsilon(\psi_\varepsilon) \to 0$
- Obtain conditions for minimax nondistinguishability i.e. $\gamma_\varepsilon \to 1$

Proposition 3.1. *Let $\pi_{\varepsilon,0}, \pi_{\varepsilon,1}$ be families of priors such that :*

$$\pi_{\varepsilon,0}(\Theta_{\varepsilon,0}) \to 1 \;\; \textit{and} \;\; \pi_{\varepsilon,1}(\Theta_{\varepsilon,1}) \to 1$$

Then, the following lower bounds hold true :

$$\begin{aligned}\beta_\varepsilon(\alpha) &\ge \beta_\varepsilon(\alpha; P_{\pi_{\varepsilon,0}}, P_{\pi_{\varepsilon,1}}) + o(1)\\ \gamma_\varepsilon &\ge \gamma_\varepsilon(P_{\pi_{\varepsilon,0}}, P_{\pi_{\varepsilon,1}}) + o(1) = 1 - \frac{1}{2}TV(P_{\pi_{\varepsilon,0}}, P_{\pi_{\varepsilon,1}}) + o(1)\end{aligned}$$

The problem that occurs often is that it is difficult to compute the total variation distance, so instead, we will use the χ^2-divergence that one can define as follows :

Definition 3.2. *Let P_0, P_1 be two probability measures on the same probability space. We define the χ^2-divergence between P_0 and P_1 as follows :*

$$|P_0 - P_1|_2^2 = \chi^2(P_0\|P_1) = \begin{cases}\infty & \textit{if } P_0 \textit{ does not dominate } P_1\\ \mathbb{E}_{P_0}\left[\frac{dP_1}{dP_0} - 1\right]^2 & \textit{otherwise}\end{cases} \tag{3.3}$$

Below, we will assume that $P_{\varepsilon,0}$ dominates all the other measures $P_{\varepsilon,\theta}$ for $\theta \in \Theta$ (This is fulfilled in the Gaussian sequence model)

Proposition 3.2 (χ^2 nondistinguishability)**.**

- *Let $|P_{\varepsilon,1} - P_{\varepsilon,0}|_2 \to 0$. Then $TV(P_{\varepsilon,0}, P_{\varepsilon,1}) \to 0$ and $\gamma_\varepsilon(P_{\varepsilon,0}, P_{\varepsilon,1}) \to 0$.*
- *If $|P_{\varepsilon,1} - P_{\varepsilon,0}|_2 = \mathcal{O}(1)$, then $\limsup TV(P_{\varepsilon,0}, P_{\varepsilon,1}) < 2$ and $\liminf \gamma_\varepsilon(P_{\varepsilon,0}, P_{\varepsilon,1}) > 0$*

Proof. Add proof □

3.2.2 Separation radius and goodness-of-fit

In Goodness-of-Fit problems (i.e. $\Theta_{\varepsilon,1} = \Theta_{\varepsilon,0}^C$ in Equation (3.2)), it is impossible to use directly minimax approach because :

$$\forall\varepsilon : \gamma_\varepsilon(\Theta_{\varepsilon,0}, \Theta_{\varepsilon,1}) \equiv 1$$

Therefore, we consider alternatives of the form :

$$\Theta_{\varepsilon,1} = \left\{\theta \in \Theta : d(\theta, \Theta_{\varepsilon,0}) \geq r_\varepsilon\right\}$$

Where d is a distance function and for $\theta \in \Theta : d(\theta, \Theta_{\varepsilon,0}) := \inf_{\theta_0 \in \Theta_{\varepsilon,0}} d(\theta, \theta_0)$. Now, our problem can be stated as follows : What are the radii r_ε in order to obtain minimax distinguishability $\gamma_\varepsilon \to 0$?
We will be studying this in the specific case of simple null-hypothesis of the form : $\mathcal{H}_0 : \theta \in \Theta_{\varepsilon,0} = \{\theta_0\}$.

Definition 3.3 (Critical radii). *We say that a family r_ε^* is the critical radii if it satisfies the following dichotomy :*

$$\gamma_\varepsilon \to 0 \Leftrightarrow \frac{r_\varepsilon}{r_\varepsilon^*} \to \infty$$
$$\gamma_\varepsilon \to 1 \Leftrightarrow \frac{r_\varepsilon}{r_\varepsilon^*} \to 0$$

Moreover, we say that distinguishability conditions are in the classical form if the are in the form above.

TO DO

3.2.3 Minimax properties of χ^2 type tests

Now, return to the Gaussian model in the space of sequences ℓ^2, and condiser the following hypothesis problem :

$$\mathcal{H}_0 : v = 0 \text{ versus } \mathcal{H}_1 : v \in V_\varepsilon \subseteq \ell^2$$

Definition 3.4 (χ^2-type tests). *We say that a test ψ is of χ^2-type if there exists a family $w_\varepsilon = \{w_{\varepsilon,i}\}$ of sequences such that :*

$$\psi = \psi_{\varepsilon,w_\varepsilon;T} := \mathbb{1}_{t_\varepsilon > T_\varepsilon} \text{ where } t_\varepsilon = \sum_{i\in\mathbb{N}} w_{\varepsilon,i}(x_i - 1)^2$$

In the context of the definition above, set :

$$h_\varepsilon(v) = \sum_{i\in\mathbb{N}} w_{\varepsilon,i} v_i^2;\ t_\varepsilon(v) = t_\varepsilon - h_\varepsilon(v);\ |w_\varepsilon| = \sup_{i\in\mathbb{N}} w_{\varepsilon,i}$$

Under the null hypothesis, one can note that $\mathbb{E}_0 t_\varepsilon$, moreover, one can assume the normalization $\sum w_{\varepsilon,i}^2 = \frac{1}{2}$ in order to ensure that $\mathrm{Var}(t_\varepsilon) = 1$

Lemma 3.1 (Minimax properties of χ^2-type tests). *Ingster and Suslina [2012] Assume that the family w_ε satisfies $w_{\varepsilon,i} \geq 0$ as well as the normalization above. Then :*

$$Var_v(t_\varepsilon) \leq 1 + 4|w_\varepsilon| h_\varepsilon(v) \tag{3.4}$$

Moreover, if $h_\varepsilon := \inf_{v\in V_\varepsilon} h_\varepsilon(v, w_\varepsilon) \to \infty$ then for $T_\varepsilon = \frac{h_\varepsilon}{2}$, one has $\gamma(\psi_{w_\varepsilon,T_\varepsilon}, V_\varepsilon)$. Additionally, if one assumes that $|w_\varepsilon|$, then the statistics $t_\varepsilon(v)$ are asymptotically $\mathbb{N}(0,1)$ uniformly over $v \in \ell^2$ such that $h_\varepsilon(v) = o(|w_\varepsilon|^{-1})$, and the following inequalities hold true :

$$\beta(\psi_{\varepsilon,T_\alpha}, V_\varepsilon) \leq \Phi(T_\alpha - h_\varepsilon) + o(1),\ \gamma(\psi_{\varepsilon,T_\varepsilon}) \leq 2\Phi(-\frac{h_\varepsilon}{2}) + o(1)$$

Proof. ADD PROOF ∎

Chapter 4

Goodness-of-fit using Classifier-Accuracy tests

Maybe a small introduction to the goal of this section and the theorem that we want to prove

4.1 Setting of the problem

Definition 4.1. *Let μ, ν be two probability measures on a space $\mathcal{X}$, we define the total variation between μ and ν as follows :*

$$TV(\mu, \nu) = \sup_{E \subset \mathcal{X} measurable} |P(E) - Q(E)|$$

If μ, ν have densities f, g respectively with respect to another measure λ, then one has the equality :

$$TV(\mu, \nu) = \int |p(x) - q(x)| \, d\lambda(x)$$

Now, suppose that one has two samples X, Y of size n from two distributions $\mathbb{P}_X$ and $\mathbb{P}_Y$ respectively on some space $\mathcal{X}$ and wishes to test the hypotheses :

$$\mathcal{H}_0 : \mathbb{P}_X = \mathbb{P}_Y \text{ versus } \mathcal{H}_1 : TV(\mathbb{P}_X, \mathbb{P}_Y) \geq \varepsilon$$

for some $\varepsilon > 0$, by using machine learning classifiers. Given such hypotheses $\mathcal{H}_0, \mathcal{H}_1$ seen as families of distributions on $\mathcal{X}$, we say that a decision rule $\psi : \mathcal{X} \to \{0, 1\}$ tests the two hypothesis against each other with error at most $\delta \in \left\{0, \frac{1}{2}\right\}$ if :

$$\max_{i=0,1} \max_{P \in \mathcal{H}_i} \mathbb{P}_{S \sim P} \Big(\psi(S) \neq i \Big) \leq \delta$$

The idea of using classifiers, firstly described by Friedman [2004], consists of training a binary classifier $\mathcal{C} : \mathcal{X} \to \{0, 1\}$ on the labeled data $\cup_{i=1}^{n} \left\{ (X_i, 0), (Y_i, 0) \right\}$ and compare the samples $\mathcal{C}(X_1), \dots, \mathcal{C}(X_n)$ to $\mathcal{C}(Y_1), \dots, \mathcal{C}(Y_n)$. **ADD MORE DETAILS**
We will follow the framework described by Gerber et al. [2023] as "Classifier-Accuracy testing" (CAT) for goodness-of-fit testing :

Suppose we have a sample X of size $2n$ from an unknown distribution $\mathbb{P}_X$ that we will split into 2 halves $X = X^{\text{tr}} \sqcup X^{\text{te}}$ and another sample Y of size $2n$ from a known distribution $\mathbb{P}_Y$ which corresponds to our null hypothesis, we will split this sample in the same manner as for the previous sample into $Y = Y^{\text{tr}} \sqcup Y^{\text{te}}$. We train a classifier $\mathcal{C} : \mathcal{X} \to \{0, 1\}$ on the input $\mathcal{D}^{\text{tr}} :== \left\{ X^{\text{te}}, Y^{\text{te}} \right\}$ that aims to assign 1 to $\mathbb{P}_X$ and 0 to $\mathbb{P}_Y$. In practice, it is easier to think of $\mathcal{C}$ in terms of the separating set $S := \mathcal{C}^{-1}(\{1\})$ which is a random set whose randomness come from the splitting of our samples and potentially an external seed.

Definition 4.2. *Given two datasets $\left\{A_i\right\}_{i=1}^{a}, \left\{B_j\right\}_{j=1}^{b}$, we define the classifier-accuracy stastistic s :*

$$T_S(A, B) := \frac{1}{a} \sum_{i=1}^{a} \mathbb{1}_{A_i \in S} - \frac{1}{b} \sum_{j=1}^{b} \mathbb{1}_{B_j \in S}$$

Finally, we threshold the output $|T_S|$ of the test data $\mathcal{D}^{\text{te}} := \{X^{\text{te}}, Y^{\text{te}}\}$

The goal of this section will be two prove the following lower bound on the sample complexity of CAT testing, i.e. the minimum size of the sample for which there exists a CAT test that tests the two hypotheses against each other with probability error less than δ. This result can be summarized in the following theorem.

Theorem 4.1. *Gerber et al. [2023] Given a known distribution $\mathbb{P}_0$ following a Gaussian sequence model with mean vector in the Sobolev ellipsoid $\mathcal{E}(s, C)$, consider the following goodness-of-fit problem :*

$$\mathcal{H}_0 : \mathbb{P}_X = \mathbb{P}_0 \;\; versus \;\; \mathcal{H}_1 : TV(\mathbb{P}_X, \mathbb{P}_0) \geq \varepsilon$$

Where P_X is the distribution of our sample X. Let $n(\varepsilon, \delta, s)$ be the smallest number such that for all $n \geq n(\varepsilon, \delta, s)$ there exists a function $\psi : \mathcal{X}^n \to \{0, 1\}$ which given our sample X as input, tests between $\mathcal{H}_0$ and $\mathcal{H}_1$ with error probability at most δ for every $\mathbb{P}_X, \mathbb{P}_0 \in \mathcal{E}(s, C)$. Then :

$$n(\varepsilon, \delta, s) \gtrsim \frac{\sqrt{\log(\frac{1}{\delta})}}{\varepsilon^{\frac{4s+1}{2s}}} + \frac{\log(\frac{1}{\delta})}{\varepsilon^2}$$

4.1.1 The Sobolev ellipsoid and Gaussian sequence model

In this project, we will restrict ourselves to a very specific class of distributions, the Gaussian sequence model with mean vector in the Sobolev ellipsoid that we will define below.

Definition 4.3. *Let $s > 0, C > 0$. We define the Sobolev ellipsoid $\mathcal{E}(s, C)$ as follows :*

$$\mathcal{E}(s, C) := \left\{\theta \in \mathbb{R}^{\mathbb{N}} : \sum_{j=1}^{\infty} j^{2s}\theta_j^2 \leq C\right\}$$

For a sequence $\theta \in \mathcal{E}(s, C)$, we define $\mu_\theta = \bigotimes_{j=1}^{\infty} \mathcal{N}(\theta_j, 1)$. The set of Gaussian sequence model distributions is hence defined as :

$$\mathcal{P}_G(s, C_G) := \{\mu_\theta; \theta \in \mathcal{E}(s, C_G)\}$$

Intuition about the choice of this model

4.2 Preliminary results

The hard part of this kind of frameworks consists in finiding a good separating set $S \subset \mathcal{X}$, but we firstly need to define what "good" means.

Definition 4.4. *Let $S \subseteq \mathcal{X}$ be a measurable set, and μ, ν be two probability measures on $\mathcal{X}$. We define the separation and size of S respectively as follows :*

$$sep(S) = \mu(S) - \nu(S) \;\; and \;\; \tau(S) = \min(\mu(S)\mu(S^c), \nu(S)\nu(S^c))$$

Lemma 4.1. *Gerber et al. [2023, Lemma 1]*
Consider the hypothesis testing problem $\mathcal{H}_0 : p = q$ versus any other alternative $\mathcal{H}_1$. Suppose that the learner has constructed a separating set S such that :

$$\forall \mu, \nu \in \mathcal{H}_1 : |sep(S)| = |\mu(S) - \nu(S)| \geq \underline{sep} \;\; and \;\; \forall \mu, \nu \in \mathcal{H}_0 \cup \mathcal{H}_1 : \min(\mu(S)\mu(S^c), \nu(S)\nu(S^c)) \leq \bar{\tau}$$

Then, using only the knowledge of $\bar{\tau}$, the classifier-accuracy test with samples of size n from both p and q and an appropriate threshold achieves type-I and type-II errors at most δ provided that :

$$n \geq c\frac{\log(\frac{1}{\delta})}{\underline{sep}}\left(1 + \frac{\bar{\tau}}{\underline{sep}}\right)$$

for a large enough constant $c > 0$

Remark. *With Lemma 4.1, one can imagine that a separating can be referred to as "good" if $|sep(S)|$ is large under H_1, and $\tau(S)$ is small under both H_0 and $\mathcal{H}_1$ with probability $1 - \delta$*

In order to prove lemma 4.1, we will need a prior result on the Bernstein concentration of the classifier accuracy statistics.

Lemma 4.2. *Gerber et al. [2023, Lemma 8]*
Suppose $A_1, A_2, \ldots, A_n \overset{iid}{\sim}$ Ber(p) and $B_1, B_2, \ldots, B_m \overset{iid}{\sim}$ Ber(q). Let $\tau = \min(p(1-p), q(1-q))$, and let $\bar{A}, \bar{B}$ be their empirical means, respectively. There exists a universal constant $c > 0$ such that :

$$\mathbb{P}\left(\left|\bar{A} - \bar{B}\right| \leq \frac{1}{2}\left|p - q\right| - \frac{1}{2}\sqrt{\frac{c\log(1/\delta)\tau}{\min(n,m)}} - \frac{1}{2}\frac{c\log(1/\delta)}{\min(n,m)} \right) \leq \delta,$$
$$\mathbb{P}\left(\left|\bar{A} - \bar{B}\right| \geq 2\left|p - q\right| + 2\sqrt{\frac{c\log(1/\delta)\tau}{\min(n,m)}} + 2\frac{c\log(1/\delta)}{\min(n,m)} \right) \leq \delta$$

Proof of lemma 4.1. Using n test samples from (X, Y), consider the following classifier-accuracy test: we accept $\mathcal{H}_0$ if

$$\left| \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}_{X_i \in S} - \mathbb{1}_{Y_i \in S} \right| \leq \sqrt{\frac{c\bar{\tau}\log(1/\delta)}{n}} + \frac{c\log(1/\delta)}{n}$$

and reject the null hypothesis otherwise. Above, $c > 0$ is a large absolut constant, and note that the threshold only relies on the knowledge of $\bar{\tau}, n$ and δ
To analyse type-I and type-II errors , first, assume that $\mathcal{H}_0$ holds. Since sep$(S) = 0$ under the latter hypothesis, the second statement of Lemma 4.2 implies that we accept $\mathcal{H}_0$ with probability at least $1 - \frac{\delta}{2}$ if c is large enough. Moreover, if $\mathcal{H}_1$ holds, with probability at least $1-\frac{\delta}{2}$, the first statement of Lemma 4.2 implies that :

$$\left| \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}_{X_i \in S} - \mathbb{1}_{Y_i \in S} \right| \geq |\underline{\text{sep}}| + \left(\sqrt{\frac{c\bar{\tau}\log(1/\delta)}{n}} + \frac{c\log(1/\delta)}{n}\right)$$

By the assumed lower bound $n \geq c\frac{\log(\frac{1}{\delta})}{\underline{\text{sep}}}\left(1 + \frac{\bar{\tau}}{\underline{\text{sep}}}\right)$, we will reject $\mathcal{H}_0$ as desired. ☐

Proposition 4.1. *Gerber et al. [2023, Prop 1] Let $\mathbb{P}$ and $\mathbb{Q}$ be two probablity measures on a probability space $\mathcal{X}$, and let $\mathcal{C}$ be a possibly randomized classifier. Then the following inequality is true :*

$$TV(\mathbb{P}, \mathbb{Q}) \leq 2\sup\left\{\mathbb{P}(\mathcal{C}(X) = 0) - \mathbb{Q}(\mathcal{C}(X) = 0) \text{ over } X \text{ satisfying } \mathbb{P}(\mathcal{C}(X) = 0) = \mathbb{Q}(\mathcal{C}(X) = 1)\right\}$$

In the inequality above, the constant 2 is tight.

In order to prove Proposition 4.1 we will need the following lemma :

Lemma 4.3. *Let μ be a non-negative measure on a measurable space $\mathcal{X}$, and let $a, b : \mathcal{X} \to \mathbb{R}_+$ such that $\int a(x)d\mu(x) > 0$ and $\forall x \in \mathcal{X} : b(x) = 0 \implies a(x) = 0$. Then the following holds :*

$$\inf_{x \in supp(\mu)} \frac{a(x)}{b(x)} \leq \frac{\int a(x)d\mu(x)}{\int b(x)d\mu(x)} \leq \sup_{x \in supp(\mu)} \frac{a(x)}{b(x)}$$

Proof of Proposition 4.1. Let p, q be the densities of $\mathbb{P}, \mathbb{Q}$ with respect to a common dominated measure (e.g. $\mathbb{P} + \mathbb{Q}$), and define $E := \{p(x) > q(x)\}$, then $TV(\mathbb{P}, \mathbb{Q}) = \mathbb{P}(E) - \mathbb{Q}(E) \geq 0$. Without loss of generality, assume that $\mathbb{P}(E) + Q(\mathbb{E}) \geq 1$, and for $t \in [0, 1]$, define :

$$E_t := \left\{ . \frac{p(x) - q(x)}{p(x) + q(x)} \geq t \right\}$$

Then, the map $t \mapsto \mathbb{P}(E_t) + \mathbb{Q}(E_t)$ is non-increasing and left-continuous, and note that $E_0 = E$ and $E_1 = \emptyset$. Hence, the maximum $t^* := \max\left\{t \in [0,1] : \mathbb{P}(E_t) + \mathbb{Q}(E_t) \geq 1\right\}$ exists. Now, define the randomized classifier $\mathcal{C}$ as follows :

$$\mathcal{C}(x) := \begin{cases} 0 & \text{if } x \in E_{(t^*)^+} \\ 1 & \text{if } x \notin E_{t^*} \\ \text{Ber}(x) & \text{if } x \in E_{t^*} \backslash E_{(t^*)^+} \end{cases}$$

Where :

$$E_{(t^*)^+} := \cap_{t > t^*} E_t \text{ and } r = \frac{1 - \mathbb{P}(E_{(t^*)^+}) - \mathbb{Q}(E_{(t^*)^+})}{\mathbb{P}(E_{t^*}) + \mathbb{Q}(E_{t^*}) - \mathbb{P}(E_{(t^*)^+}) - \mathbb{Q}(E_{(t^*)^+})} \in [0, 1]$$

Note that this classifier is balanced since $\mathbb{P}(\mathcal{C}(X)=0)+\mathbb{Q}(\mathcal{C}(X)=0)=1$.
Now, for $t \in [0,1]$, define :

$$f(t) := \begin{cases} \frac{\mathbb{P}(E_t)-\mathbb{Q}(E_t)}{\mathbb{P}(E_t)+\mathbb{Q}(E_t)} & \text{if } \mathbb{P}(E_t)+\mathbb{Q}(E_t)>0 \\ 1 & \text{otherwise} \end{cases}$$

Now, let $0 \le t \le s \le 1$, we want to show that $f(t) \le f(s)$. Without loss of generality, assume that $f(s) < 1$ and that $\mathbb{P}(E_t)+\mathbb{Q}(E_t)>0$. Notice also that (it can be proven using simple algebraic manipulations):

$$f(t) \le f(s) \Leftrightarrow \frac{\int_{E_t \setminus E_s} p-q}{\int_{E_t \setminus E_s} p+q} \le \frac{\int_{E_s} p-q}{\int_{E_s} p+q}$$

However, this inequality follows from Lemma 4.3. In fact :

$$Addmoredetails$$

Thus, it holds that :

$$\frac{\mathbb{P}\big(\mathcal{C}(X)=0\big)-\mathbb{Q}\big(\mathcal{C}(X)=0\big)}{\mathbb{P}\big(\mathcal{C}(X)=0\big)+\mathbb{Q}\big(\mathcal{C}(X)=0\big)} \ge f(t^*) \ge f(0) = \frac{\mathbb{P}(E)-\mathbb{Q}(E)}{\mathbb{P}(E)+\mathbb{Q}(E)}$$

Plugging in $\mathbb{P}\big(\mathcal{C}(X)=0\big)+\mathbb{Q}\big(\mathcal{C}(X)=0\big)=1$ and $\mathbb{P}(E)+\mathbb{Q}(E) \le 2$ yields the final result. For the tightness, consider $p(x)=\mathbb{1}_{[0,1], q(x)=(1+\varepsilon)\mathbb{1}_{[0,\frac{1}{1+\varepsilon}]}, \mathcal{C}(x)=\mathbb{1}_{(\frac{1}{2+\varepsilon},1]}}$ and make $\varepsilon \to 0$. □

Lemma 4.4 (Lipschitz concentration of Gaussians). *Let Q be a d-dimensional standard gaussian, and let $f : \mathbb{R}^d \to \mathbb{R}$ be L-Lipschitz. Then $f(Q)$ is sub-Gaussian with variance proxy L^2*

Proof of Lemma 4.4. Add proof, log-Sobolev inequality □

Lemma 4.5. *Let $a, b \in \mathbb{R}$, and let Z be a standard normal. Then :*

$$\mathbb{E}\Phi(aZ+b) = \Phi\left(\frac{b}{\sqrt{1+a^2}}\right)$$

4.3 Performance of the CAT test in the Gaussian case

Suppose one have two samples X, Y of size $n \in \mathbb{N}$ from $\mu_{\theta^X} := \bigotimes_{j=1}^{\infty} \mathcal{N}(\theta_j^X, 1)$ and μ_{θ^Y}, respectively and such that μ^X, μ^Y have bounded Sobolev norm. Moreover, assume that $TV(\mu_{\theta^X}, \mu_{\theta^Y}) \ge \varepsilon > 0$, and let $\hat{\theta}^X, \hat{\theta}^Y$ denote the empirical means from the two samples respectively. Define the following statistic :

$$T(Z) = 2\sum_{j=1}^{J}\left(\hat{\theta}_j^X - \hat{\theta}_j^Y\right)\left(Z_j - \frac{\hat{\theta}_j^X + \hat{\theta}_j^Y}{2}\right) \tag{4.1}$$

for some $J \in \mathbb{N}$ to be specified. We construct the separating set as follows :

$$\hat{S} := \left\{Z \in \mathbb{R}^{\mathbb{N}} : T(Z) \ge 0\right\} \tag{4.2}$$

The performance of this construction of separating set is summarized in the statement of the following result :

Proposition 4.2. *Gerber et al. [2023] There exists universal constants c, c' such that for $J = \lfloor c\varepsilon^{-\frac{1}{s}} \rfloor$ the following inequality holds provided $n \gtrsim \frac{1}{c'} n(\varepsilon, \delta)$:*

$$\mathbb{P}\left(\mu_{\theta^X}(\hat{S}) - \mu_{\theta^Y}(\hat{S}) \ge c' \cdot \min\left(\sqrt{n\varepsilon^{\frac{1}{s}}}, \frac{1}{\varepsilon}\right) \cdot \varepsilon^2\right) \ge 1-\delta \tag{4.3}$$

In order to prove this proposition, we will need few lemmas and prior results.

Lemma 4.6. *Let $sep(\hat{S}) = \mu_{\theta^X}(\hat{S}) - \mu_{\theta^Y}(\hat{S})$ as defined above. There exists universal constants $c_i > 0$, $i = 1,2,\cdots,5$ such that for $J = \lfloor c_1 \varepsilon^{-\frac{1}{s}} \rfloor$ and for $t \geq 0$, one has :*

$$\mathbb{E}[sep(\hat{S})] + \frac{c_2}{\sqrt{n}} \geq \frac{c_3 \varepsilon^2}{\varepsilon + \sqrt{\frac{J}{n}}} \tag{4.4}$$

$$\mathbb{P}\left(\left| sep(\hat{S}) - \mathbb{E}sep(\hat{S}) \right| \geq t + \frac{c_4}{\sqrt{n}} \right) \leq 2\exp(-c_5 n t^2) \tag{4.5}$$

Proof of Lemma 4.6. Let $\|\cdot\|, \langle \cdot, \cdot \rangle$ for the ℓ^2 norm/inner product restricted to the first J coordinates. Notice that given $\hat{\theta}^X, \hat{\theta}^Y, T(\theta)$ is a Gaussian random variable with :

$$\mathbb{E}T(\theta) = \|\hat{\theta}^Y - \theta\|^2 - \|\hat{\theta}^X - \theta\|^2 \text{ and } \mathrm{Var}(T) = 4\|\hat{\theta}^X - \hat{\theta}^Y\|^2$$

Now, define the random vectors U, V as follows :

$$U = \{\hat{\theta}^X_j - \hat{\theta}^Y_j\}_{j=1}^J$$
$$V = \{\hat{\theta}^X_j + \hat{\theta}^Y_j\}_{j=1}^J$$

One can easily see that these two vectors are independent, Gaussians with covariance matrix $\frac{2}{n} I_J$ and means equal to the first J coordinates of $\theta^X \mp \theta^Y$ respectively. Let Φ be the cumulative distribution function of the standard Gaussian and ϕ be its density. It can be shown that the separation can be written as :

$$\mathrm{sep}(\hat{S}) = f(\hat{\theta}^X) - f(\hat{\theta}^Y)$$

Where f is defined as follows :

$$f(\theta) = \Phi\left(\frac{\|\hat{\theta}^Y - \theta\|^2 - \|\hat{\theta}^X - \theta\|^2}{2\|\hat{\theta}^X - \hat{\theta}^Y\|^2} \right) = \Phi\left(-\frac{1}{2}\left\langle V, \frac{U}{\|U\|} \right\rangle + \left\langle \theta, \frac{U}{\|U\|} \right\rangle \right)$$

In order to make the dependences more clear, write $g(U, V) = f(\theta^X) - f(\theta^Y)$. Now, differentiating g and using the fact that ϕ is $\frac{1}{\sqrt{2\pi e}}$-Lipschitz yields :

$$\begin{aligned}
\|\nabla_V g(U, V)\| &= \Bigg\| -\frac{1}{2}\frac{U}{\|U\|}\Bigg(\phi\Big(-\frac{1}{2}\Big\langle V, \frac{U}{\|U\|}\Big\rangle + \Big\langle \theta^X, \frac{U}{\|U\|}\Big\rangle \Big) \\
&\qquad - \phi\Big(-\frac{1}{2}\Big\langle V, \frac{U}{\|U\|}\Big\rangle + \Big\langle \theta^Y, \frac{U}{\|U\|}\Big\rangle \Big) \Bigg) \Bigg\| \\
&\leq \frac{1}{\sqrt{8\pi e}} \left| \left\langle \theta^X - \theta^Y, \frac{U}{\|U\|} \right\rangle \right| \\
&\leq \frac{C}{\sqrt{8\pi e}}
\end{aligned}$$

By Lipschitz concentration of Gaussian random variable (Lemma 4.4), one can conclude that $g - \mathbb{E}[g|U]$ is sub-Gaussian with variance proxy $\frac{C^2}{4\pi e n}$. Now, we study the concentration of $\mathbb{E}[g|U]$. First, note that :

$$-\frac{1}{2}\left\langle V, \frac{U}{\|U\|} \right\rangle + \left\langle \theta, \frac{U}{\|U\|} \right\rangle \Bigg| U \sim \mathcal{N}\left(\left\langle \theta - \frac{1}{2}(\theta^X + \theta^Y), \frac{U}{\|U\|} \right\rangle, \frac{1}{2n} \right)$$

Then, using the independence of U, V and lemma 4.5, we obtain :

$$\begin{aligned}
\mathbb{E}[g(U, V)|U] &= \mathbb{E}[f(\theta^X) - f(\theta^Y)|U] \\
&= \Phi\left(\frac{W}{\sqrt{4 + \frac{2}{n}}} \right) - \Phi\left(-\frac{W}{\sqrt{4 + \frac{2}{n}}} \right)
\end{aligned}$$

Where $W := \langle \theta^X - \theta^Y, \frac{U}{\|U\|} \rangle$. To simplify notations, let $\varphi := \Phi\Big(\frac{\cdot}{\sqrt{4 + \frac{2}{n}}} \Big)$, by Lipschitness of Φ, we obtain that for $t \geq 0$:

$$\mathbb{P}\Big(\Big| \varphi(W) - \mathbb{E}\varphi(W) \Big| \geq t \Big) \leq \mathbb{P}\Big(\Big) \color{red}{Finishtheproof}$$

Lemma 4.7. *W is a sub-Gaussian random variable with variance proxy $\frac{1}{2n}$*

Proof in the appendix ☐

Proof of Proposition 4.2. Add proof ☐

4.4 Sample complexity of CAT for Gaussian sequence model

Given two hypotheses H_0, H_1, our goal is to find a lower bound for the minimum achievable worst-case error. Firstly, we use the fact that for any random probability distributions P_0, P_1 with $\mathbb{P}(P_i \in H_i) = 1$ for $i = 1, 2$, one has that :

$$\min_{\psi} \max_{i=0,1} \sup_{P \in H_1} \mathbb{P}_{S\sim P}\big(\psi(S) \neq i\big) \geq \frac{1}{2}\big(1 - TV(\mathbb{E}P_0, \mathbb{E}P_1)\big) \tag{4.6}$$

Where $\mathbb{E}P$ denotes the mixture of P. Also, note that deriving a lower bound of order δ on the minimax error is equivalent to finding mixtures $\mathbb{E}P_0, \mathbb{E}P_1$ such that $1 - TV(\mathbb{E}P_0, \mathbb{E}P_1)$ is of order δ (is $\Omega(\delta)$)

Lemma 4.8. *Let μ, ν be two probability measures, then :*

$$1 - TV(\mu, \nu) \geq \frac{1}{2}e^{-KL(\mu\|\nu)} \geq \frac{1}{2\big(1 + \chi^2(\mu\|\nu)\big)}$$

Where $KL(\cdot\|\cdot)$ and $\chi^2(\cdot\|\cdot)$ denote the Kullback-Leibler and χ^2 divergences respectively.

Proof of Lemma 4.8. We will show that each one of the two inequalities hold true using Jensen inequality.

- Equivalently, we will show that :

  $$\int \log\Big(\frac{d\mu}{d\nu}\Big)d\mu = KL(\mu\|\nu) \leq \log(1 + \chi^2(\mu\|\nu)) = \log\Big(\int \frac{d\mu}{d\nu}d\mu\Big)$$

  Which is a direct consequence of Jensen inequality (since $\log$ is concave)

- Instead of the inequality above, one will show that :

  $$TV(\mu, \nu) \leq \sqrt{1 - e^{-KL(\mu\|\nu)}}$$

  First, notice that :

  $$TV(\mu, \nu) = 1 - \int (d\mu \wedge d\nu) = \int (d\mu \vee d\nu) - 1$$

  This implies that :

  $$\begin{aligned}
  1 - TV(\mu, \nu)^2 &= \big(1 - TV(\mu, \nu)\big)\big(1 + TV(\mu, \nu)\big) \\
  &= \int (d\mu \wedge d\nu) \int (d\mu \vee d\nu) \\
  &\overset{\text{C-S}}{\geq} \Big(\int \sqrt{fg}\Big)^2 \quad \text{(Where } f, g \text{ are the densities)} \\
  &= \exp\Big(2\log\Big(\int \sqrt{\frac{d\nu}{d\mu}}d\mu\Big)\Big) \\
  &= \exp\Big(2\log \mathbb{E}_\mu \Big(\sqrt{\frac{d\mu}{d\nu}}\Big)^{-1}\Big) \\
  &\geq \exp\Big(-\mathbb{E}_\mu \log\Big(\frac{d\mu}{d\nu}\Big)\Big) = \exp\Big(-KL(\mu\|\nu)\Big)
  \end{aligned}$$

  And finally, one can conclude using the following inequality : $\sqrt{1-x} \leq 1 - \frac{x}{2}$.

Combining the two points above yields the inequality. ☐

Now that we have all the necessary tools, we will prove the main theorem of this chapter. We will split this proof into many parts :

Proposition 4.3. *There exists a constant K_0 such that for every ε small enough,*

$$n(\varepsilon, \delta, \mathcal{P}_G(s, C_G)) \geq K_0 \frac{log(1/\delta)}{\varepsilon^2} \tag{4.7}$$

Proof of Theorem 4.1. Gerber et al. [2023] For a vector $\eta \in \{-1, 1\}^{\mathbb{N}}$, define the following probability measure :

$$\mathbb{P}_\eta := \bigotimes_{j=1}^{J} \mathcal{N}(\eta_j c_1 \varepsilon^{\frac{2s+1}{2s}}, 1) \otimes \bigotimes_{j>J} \mathcal{N}(0, 1) \tag{4.8}$$

Where $J = \lfloor c_2 \varepsilon^{-\frac{1}{s}} \rfloor$. And let $\eta_1, \eta_2, \cdots$ be iid uniform in $\{-1, 1\}$ and γ_η be the mean vector of $\mathbb{P}_\eta$. One can see that for any such η :

$$\|\gamma_\eta\|_s^2 = \sum_{j=1}^{\infty} j^{2s} \gamma_{\eta,j}^2 = \sum_{j=1}^{J} j^{2s} c^2 \varepsilon^{\frac{2s+1}{s}} \leq c_1^2 \varepsilon^{\frac{2s+1}{s}} \left(2 c_2 \varepsilon^{-\frac{1}{s}}\right)^{2s+1} \asymp c_1^2 c_2^{2s+1}$$

$$\|\gamma_\eta\|^2 = \sum_{j=1}^{\infty} \gamma_{\eta,j}^2 \asymp c_1^2 c_2 \varepsilon^2$$

Then for $C_G > 0$ (Size of Sobolev's ellipsoid) fixed, one can choose c_1, c_2 independently of ε such that $\mathbb{P}_0, \mathbb{P}_\eta \in \mathcal{P}_G(s, C_G)$ almost surely and $\|\gamma_\eta\| = 10\varepsilon$. Then, for $\varepsilon \leq \frac{1}{10}$, one has :

$$TV(\mathbb{P}_0, \mathbb{P}_\eta) = 2\Phi\left(\frac{\|\gamma_\eta\|}{2}\right) - 1 \geq \varepsilon$$

Now, take $P_0 = \mathbb{P}_0^{\otimes 2n}, P_1 = \mathbb{P}_1^{\otimes n} \otimes \mathbb{P}_0^{\otimes n}$, then :

$$KL(P_0 \| P_1) = nKL(\mathbb{P}_0 \| \mathbb{P}_1) = n c_2 \varepsilon^{-\frac{1}{s}} \frac{(c_1 \varepsilon^{\frac{2s+1}{2s}})^2}{2} = Kn\varepsilon^2$$

Using Lemma 4.8 and Equation (4.6), we get the minimax error is at least :

$$\frac{1}{2}\big(1 - TV(P_0, P_1)\big) \geq \frac{1}{4} \exp(-KL(P_0 \| P_1)) = \frac{1}{4} \exp(-Kn\varepsilon^2)$$

For this error to be less than δ, it is necessary to have :

$$\frac{1}{4}\frac{1}{4} \exp(-Kn\varepsilon^2) \leq \delta \Leftrightarrow n \geq \frac{\log(1/(4\delta))}{K\varepsilon^2}$$

Hence $n \gtrsim \frac{\log(1/\delta)}{\varepsilon^2}$ as claimed. □

Proposition 4.4. *There exists a constant $K_1 > 0$ such that for every ε small enough :*

$$n(\varepsilon, \delta, \mathcal{P}_G(s, C_G)) \geq K_1 \frac{\sqrt{\log(1/\delta)}}{\varepsilon^{\frac{4s+1}{2s}}} \tag{4.9}$$

Chapter 5

Conclusion

Appendix

5.1 Some omitted proofs

Proof of Lemma 4.3. We will prove the LHS inequality, the RHS can be proven in the exact same way :

$$\begin{aligned}\int a(x)d\mu(x) &= \int \frac{a(x)}{b(x)}b(x)d\mu(x)\\ &\geq \Big(\inf_{x\in spt(\mu)} \frac{a(x)}{b(x)}\Big)\int b(x)d\mu(x)\end{aligned}$$

The last inequality is implied by the fact that $\forall x \in \mathcal{X} : b(x) \geq 0$. $\square$

Proof of Lemma 4.5. Let Z' be a standard Gaussian independent of Z, then :

$$\mathbb{E}\Phi(aZ+b) = \mathbb{P}(aZ+b \geq Z') = \mathbb{P}\Big(\frac{Z'-aZ}{\sqrt{1+a^2}} \leq \frac{b}{\sqrt{1+a^2}}\Big) = \Phi(\frac{b}{\sqrt{1+a^2}})$$

The last equality is given by the fact that $\frac{Z'-aZ}{\sqrt{1+a^2}} \sim \mathcal{N}(0,1)$. $\square$

Proof of Lemma 4.7. Let $\tau := \theta^X - \theta^Y$ and $\sigma^2\frac{1}{2n}$ and let Q be a standard Gaussian random vector such that :

$$W = \Big\langle \tau, \frac{\tau+\sigma Q}{\|\tau+\sigma Q\|}\Big\rangle \text{ in distribution}$$

Then, one can decompose W as follows :

$$\Big\langle \tau, \frac{\tau+\sigma Q}{\|\tau+\sigma Q\|}\Big\rangle = \Big\langle \frac{\tau}{\mathbb{E}\big(\|\tau\|\big)}, ss\Big\rangle$$

Correct the proof $\square$

5.1.1 Proof of Lipschitz concentration for Gaussians

Firstly, note that we can restrict ourselves to smooth Lipschitz functions. Explicitely, if f is an L-Lipschitz function, one can consider its mollification $f_\varepsilon = f * \eta_\varepsilon$ where η is the standard mollifier. Moreover f_ε is smooth and converges uniformly on compact sets to f as $\varepsilon \to 0$. Now, we want to show the sub-Gaussianity of $f(Z)$ where $f : \mathbb{R}^d \to \mathbb{R}$ is an L-Lipschitz smooth function and Z is a standard Gaussian. This can be done using the Gaussian log-Sobolev inequality, which we will state in the following form :

Let γ_n denote the standard Gaussian measure on $\mathbb{R}^n$, that is, the law of $X \sim \mathbb{N}(0, I_n)$. We write $\mathbb{E}$ for expectation under γ_n and $\|\cdot\|_2$ for the Euclidean norm. For a nonnegative measurable function h with $\mathbb{E}[h] < \infty$ and $\mathbb{E}[h\log h] < \infty$, the *entropy of h relative to γ_n* is

$$\mathrm{Ent}_{\gamma_n}(h) := \mathbb{E}[h\log h] - \mathbb{E}[h]\log\mathbb{E}[h],$$

with the convention $0\log 0 = 0$.

Lemma 5.1 (Gaussian logarithmic Sobolev inequality)**.** *For every smooth function $g\colon \mathbb{R}^n \to \mathbb{R}$ with $g, \nabla g \in L^2(\gamma_n)$,*

$$Ent_{\gamma_n}(g^2) \leq 2\,\mathbb{E}\big[\,\|\nabla g\|_2^2\,\big].$$

Lemma 5.2 (Herbst argument)**.** *Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be smooth with $\|\nabla f(x)\|_2 \le L$ for every $x \in \mathbb{R}^n$. Then for every $\lambda \in \mathbb{R}$,*

$$\mathbb{E}\Big[e^{\lambda\,(f(X)-\mathbb{E}f(X))}\Big] \le \exp\left(\frac{\lambda^2 L^2}{2}\right).$$

Lemma 5.3 (Chernoff bound for sub-Gaussian variables)**.** *Let Y be a real-valued random variable satisfying $\mathbb{E}[e^{\lambda Y}] \le \exp\big(\lambda^2\sigma^2/2\big)$ for all $\lambda > 0$. Then for every $t > 0$,*

$$\mathbb{P}(Y \ge t) \le \exp\left(-\frac{t^2}{2\sigma^2}\right).$$

Theorem 5.1 (Gaussian Lipschitz concentration)**.** *Let $X \sim \mathbb{N}(0, I_n)$ and let $f\colon \mathbb{R}^n \to \mathbb{R}$ be L-Lipschitz. Then for every $t > 0$,*

$$\mathbb{P}\big(f(X) - \mathbb{E}f(X) \ge t\big) \le \exp\left(-\frac{t^2}{2L^2}\right),$$

and consequently

$$\mathbb{P}\big(|f(X) - \mathbb{E}f(X)| \ge t\big) \le 2\exp\left(-\frac{t^2}{2L^2}\right).$$

The bound is independent of the dimension n.

Proof. By the smoothing reduction (i.e. the reduction to smooth Lipschitz functions) it suffices to prove the inequality under the additional assumption that f is smooth and satisfies $\|\nabla f(x)\|_2 \le L$ for every $x \in \mathbb{R}^n$, since the tail probability in the conclusion is upper semicontinuous with respect to the pointwise convergence $f_k \to f$.

Define the centered random variable

$$Y := f(X) - \mathbb{E}f(X),$$

so that $\mathbb{E}Y = 0$. The Herbst argument (Lemma 5.2), whose proof relies on the Gaussian logarithmic Sobolev inequality (Lemma 5.1), asserts that

$$\mathbb{E}\big[\,e^{\lambda Y}\,\big] \le \exp\left(\frac{\lambda^2 L^2}{2}\right) \qquad \text{for every } \lambda \in \mathbb{R}.$$

In other words, Y is sub-Gaussian with variance proxy $\sigma^2 = L^2$.

Applying the Chernoff bound (Lemma 5.3) with $\sigma^2 = L^2$ then yields, for every $t > 0$,

$$\mathbb{P}\big(f(X) - \mathbb{E}f(X) \ge t\big) = \mathbb{P}(Y \ge t) \le \exp\left(-\frac{t^2}{2L^2}\right),$$

which establishes the upper-tail bound.

For the lower tail, observe that $-f$ is also L-Lipschitz, and that $\mathbb{E}(-f)(X) = -\mathbb{E}f(X)$. Applying the upper-tail bound just established to $-f$ in place of f gives

$$\mathbb{P}\big(f(X) - \mathbb{E}f(X) \le -t\big) = \mathbb{P}\big((-f)(X) - \mathbb{E}(-f)(X) \ge t\big) \le \exp\left(-\frac{t^2}{2L^2}\right).$$

Combining the two tail bounds via a union bound,

$$\mathbb{P}\big(|f(X) - \mathbb{E}f(X)| \ge t\big) \le 2\exp\left(-\frac{t^2}{2L^2}\right).$$

None of the steps above involved any dependence on the dimension n, so the estimate is dimension-free. $\square$

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